
AgileFlow

Project Management System

V 1.0

API Contract

■ 1. Authentication & Authorization

Method	Endpoint	Auth	Request Body (key fields)	Response (key fields)	Roles
POST	/api/auth/register/	None	username, email, password, role	201: user_id, message 400: errors	All
POST	/api/auth/login/	None	email, password	200: access_token, refresh_token, user 401: error	All
POST	/api/auth/logout/	JWT Bearer	refresh_token	200: message 401: unauthorized	All
POST	/api/auth/token/refresh/	None	refresh_token	200: access_token 401: invalid token	All
POST	/api/auth/password/change/	JWT Bearer	old_password, new_password	200: message 400: errors	All
POST	/api/auth/password/reset/	None	email	200: message (email sent) 404: not found	All

■ 2. User Management

Method	Endpoint	Auth	Request Body (key fields)	Response (key fields)	Roles
GET	/api/users/	JWT Bearer	— (query: role, search, page)	200: [{id, username, email, role, is_active}] paginated	Admin
POST	/api/users/	JWT Bearer	username, email, password, role, department	201: user object 400: validation errors	Admin
GET	/api/users/{id}/	JWT Bearer	—	200: full user profile 404: not found	Admin
PUT	/api/users/{id}/	JWT Bearer	username, email, role, department, is_active	200: updated user 403: forbidden	Admin
PATCH	/api/users/{id}/	JWT Bearer	Any updatable field(s)	200: updated user 400: errors	Admin
DELETE	/api/users/{id}/	JWT Bearer	—	204: no content 404: not found	Admin
GET	/api/users/me/	JWT Bearer	—	200: own profile	All
PATCH	/api/users/me/	JWT Bearer	username, avatar, bio	200: updated profile	All
GET	/api/users/{id}/projects/	JWT Bearer	—	200: [{project_id, name, role_in_project}]	Admin

■ 3. Project Management

Method	Endpoint	Auth	Request Body (key fields)	Response (key fields)	Roles
GET	/api/projects/	JWT Bearer	— (query: status, search, page)	200: paginated project list	All
POST	/api/projects/	JWT Bearer	name, description, start_date, end_date, methodology	201: project object 400: validation errors	Admin
GET	/api/projects/{id}/	JWT Bearer	—	200: full project detail 404: not found	All
PUT	/api/projects/{id}/	JWT Bearer	name, description, status, start_date, end_date	200: updated project 403: forbidden	Admin · Project Manager
PATCH	/api/projects/{id}/	JWT Bearer	Any updatable field(s)	200: updated project 400: errors	Admin · Project Manager
DELETE	/api/projects/{id}/	JWT Bearer	—	204: no content 403: forbidden	Admin
POST	/api/projects/{id}/assign-manager/	JWT Bearer	manager_id	200: {project_id, manager} 404: user not found	Admin
GET	/api/projects/{id}/members/	JWT Bearer	—	200: [{user_id, username, role}]	Admin · Project Manager
POST	/api/projects/{id}/members/	JWT Bearer	user_id, role	201: membership 400: already member	Admin · Project Manager
DELETE	/api/projects/{id}/members/{uid}/	JWT Bearer	—	204: removed 404: not found	Admin · Project Manager
GET	/api/projects/{id}/stats/	JWT Bearer	—	200: {total_tasks, completed, in_progress, overdue}	Admin · Project Manager

■ 4. Task Management

Method	Endpoint	Auth	Request Body (key fields)	Response (key fields)	Roles
GET	/api/projects/{pid}/tasks/	JWT Bearer	— (query: status, priority, assignee, page)	200: paginated task list	All
POST	/api/projects/{pid}/tasks/	JWT Bearer	title, description, assignee_id, priority, deadline, type	201: task object 400: errors	Admin · Project Manager
GET	/api/tasks/{id}/	JWT Bearer	—	200: full task detail with comments count	All
PUT	/api/tasks/{id}/	JWT Bearer	title, description, assignee_id, priority, deadline, status	200: updated task 403: forbidden	Admin · Project Manager
PATCH	/api/tasks/{id}/	JWT Bearer	Any updatable field(s)	200: updated task 400: errors	Admin · Project Manager · Team Member
DELETE	/api/tasks/{id}/	JWT Bearer	—	204: no content 403: forbidden	Admin · Project Manager
PATCH	/api/tasks/{id}/status/	JWT Bearer	status (todo in_progress review done)	200: {task_id, status, updated_at}	All
POST	/api/tasks/{id}/assign/	JWT Bearer	assignee_id	200: {task_id, assignee} 404: user not found	Admin · Project Manager
GET	/api/tasks/my-tasks/	JWT Bearer	— (query: status, priority, page)	200: tasks assigned to current user	All
GET	/api/projects/{pid}/tasks/board/	JWT Bearer	—	200: {todo:[], in_progress:[], review:[], done:[]}	All

■ 5. Team Collaboration — Comments

Method	Endpoint	Auth	Request Body (key fields)	Response (key fields)	Roles
GET	/api/tasks/{tid}/comments/	JWT Bearer	— (query: page)	200: paginated [{id, author, content, created_at}]	All
POST	/api/tasks/{tid}/comments/	JWT Bearer	content (text, max 2000 chars)	201: comment object 400: empty content	All
PUT	/api/comments/{id}/	JWT Bearer	content	200: updated comment 403: not owner	All
DELETE	/api/comments/{id}/	JWT Bearer	—	204: deleted 403: not owner/admin	All
POST	/api/comments/{id}/reactions/	JWT Bearer	reaction (like ❤️ 👍)	200: {comment_id, reactions_count}	All

■ 6. Notifications

Method	Endpoint	Auth	Request Body (key fields)	Response (key fields)	Roles
GET	/api/notifications/	JWT Bearer	— (query: is_read, page)	200: paginated [{id, type, message, is_read, created_at}]	All
PATCH	/api/notifications/{id}/read/	JWT Bearer	—	200: {id, is_read: true}	All
POST	/api/notifications/mark-all-read/	JWT Bearer	—	200: {updated_count}	All
DELETE	/api/notifications/{id}/	JWT Bearer	—	204: deleted	All
GET	/api/notifications/unread-count/	JWT Bearer	—	200: {count: N}	All

■ 7. File Management

Method	Endpoint	Auth	Request Body (key fields)	Response (key fields)	Roles
POST	/api/tasks/{tid}/files/	JWT Bearer	file (multipart/form-data), description?	201: {id, filename, size, url, uploaded_by} 413: too large	All
GET	/api/tasks/{tid}/files/	JWT Bearer	—	200: [{id, filename, size, url, uploaded_at}]	All
GET	/api/files/{id}/download/	JWT Bearer	—	200: file stream (Content-Disposition: attachment)	All
DELETE	/api/files/{id}/	JWT Bearer	—	204: deleted 403: not owner/admin	Admin · Project Manager
GET	/api/projects/{pid}/files/	JWT Bearer	— (query: page)	200: all files across project tasks	Admin · Project Manager

■ 8. Reporting & Dashboard

Method	Endpoint	Auth	Request Body (key fields)	Response (key fields)	Roles
GET	/api/dashboard/	JWT Bearer	—	200: {projects_count, tasks_count, overdue, my_tasks, activity_feed}	All
GET	/api/reports/projects/	JWT Bearer	— (query: date_from, date_to, status)	200: [{project stats per project}]	Admin · Project Manager
GET	/api/reports/tasks/	JWT Bearer	— (query: project_id, assignee_id, date_range)	200: task completion stats & breakdown	Admin · Project Manager
GET	/api/reports/users/	JWT Bearer	— (query: date_from, date_to)	200: [{user_id, tasks_completed, avg_completion_time}]	Admin
GET	/api/reports/projects/{id}/time line/	JWT Bearer	—	200: Gantt-friendly [{task, start, end, status}]	Admin · Project Manager
GET	/api/reports/export/	JWT Bearer	query: format (csv xlsx), type (tasks projects)	200: file download 400: invalid format	Admin · Project Manager

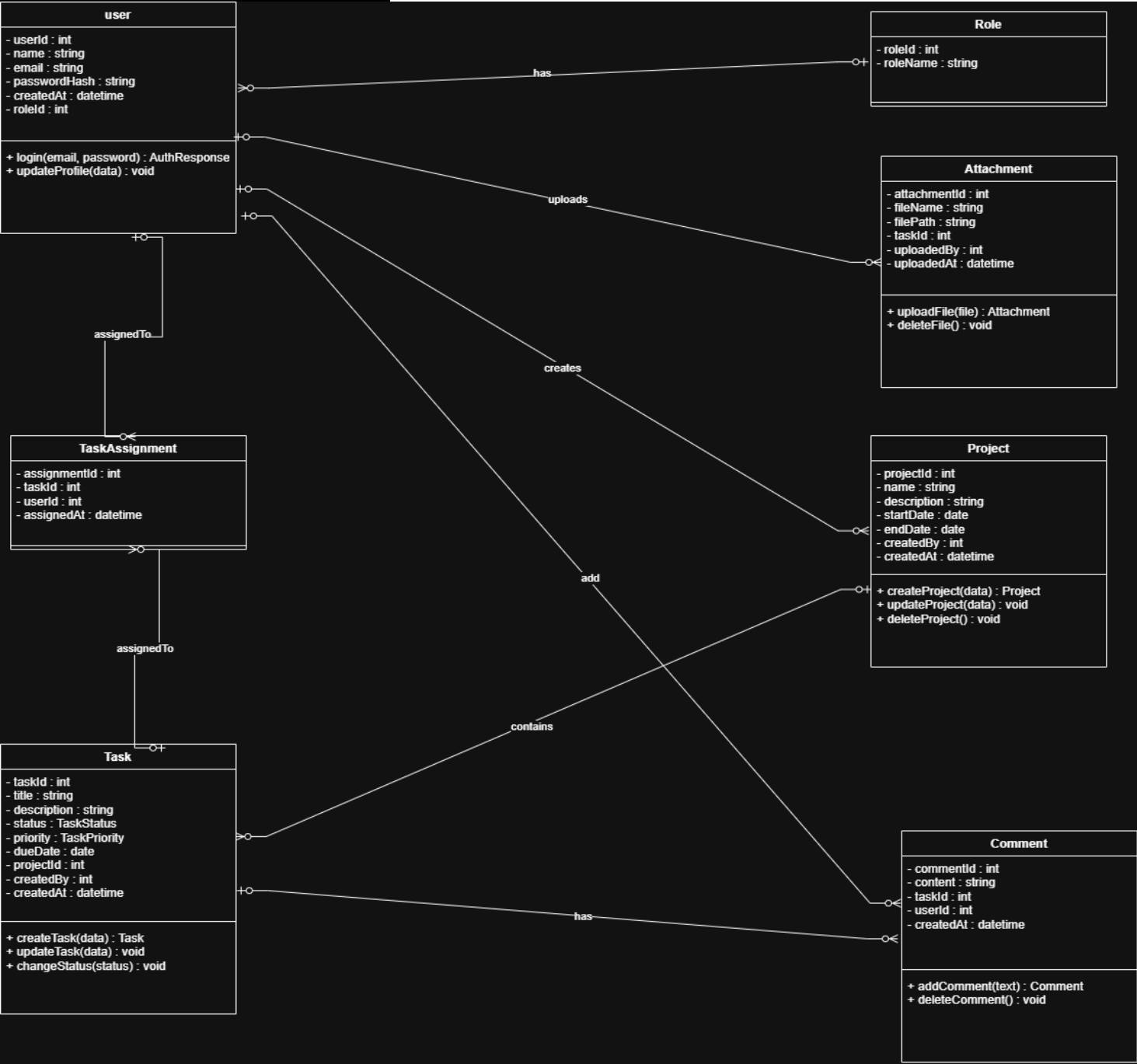
■ 9. Search & Filter

Method	Endpoint	Auth	Request Body (key fields)	Response (key fields)	Roles
GET	/api/search/	JWT Bearer	query: q (string), type (all projects tasks users), page	200: {projects:[], tasks:[], users:[], total}	All
GET	/api/projects/{id}/tasks/filter/	JWT Bearer	query: status, priority, assignee, due_before, due_after	200: filtered paginated task list	All
GET	/api/users/search/	JWT Bearer	query: q, role, department, page	200: paginated user results	Admin · Project Manager

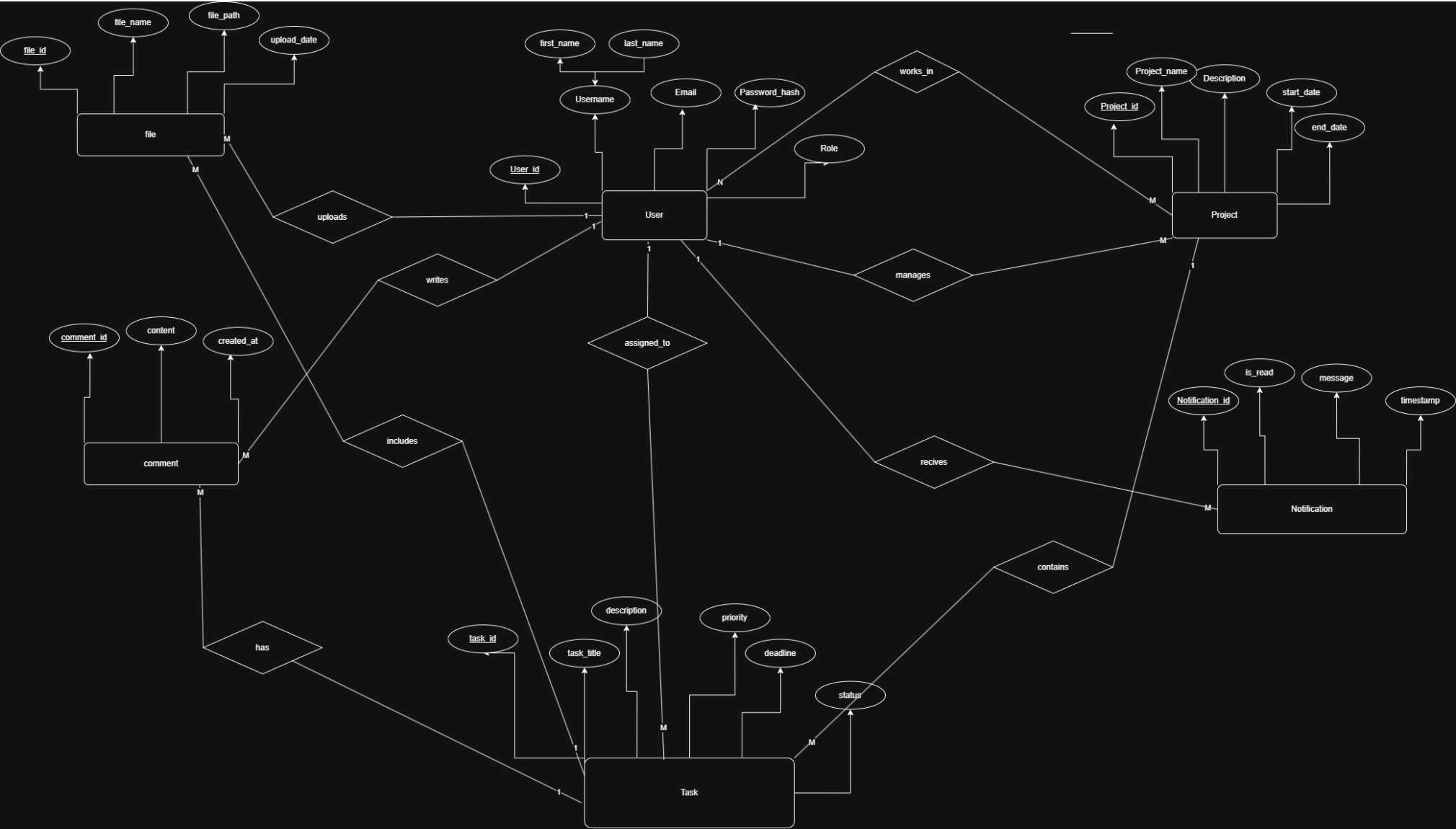
■ 10. HTTP Status Code Reference

Code	Meaning	Common Cause
200 OK	Success	Read/Update successful
201 Created	Resource created	POST success
204 No Content	Deleted	DELETE success
400 Bad Request	Validation failed	Missing/invalid fields
401 Unauthorized	No/invalid token	Missing or expired JWT
403 Forbidden	Insufficient role	Role-based access denied
404 Not Found	Resource missing	Invalid ID in URL
409 Conflict	Duplicate resource	Already exists (e.g. email)
413 Payload Too Large	File too big	Exceeds server limit
422 Unprocessable	Semantic validation error	Business rule violation
429 Too Many Requests	Rate limit hit	Throttle triggered
500 Internal Server Error	Server crash	Unhandled exception

UML Class Diagram

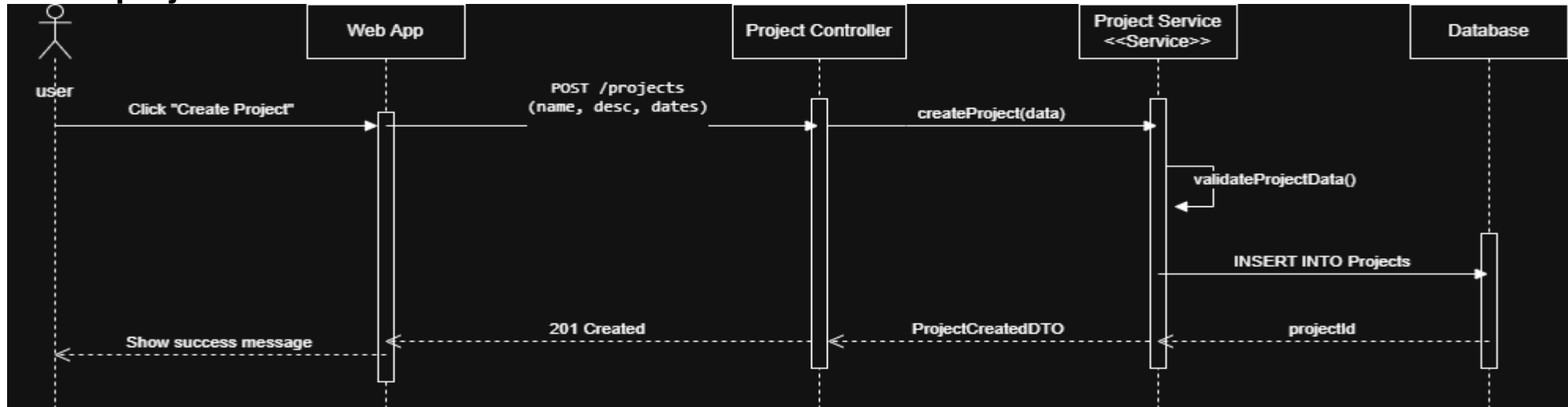


ERD Diagram

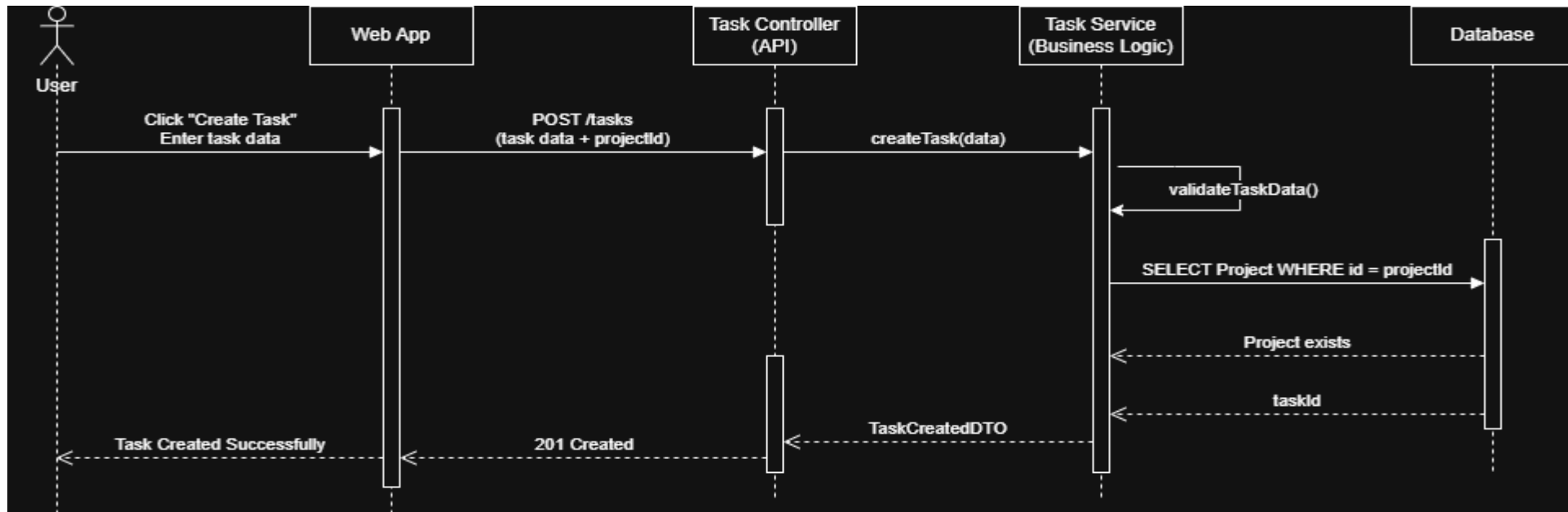


Sequence Diagram

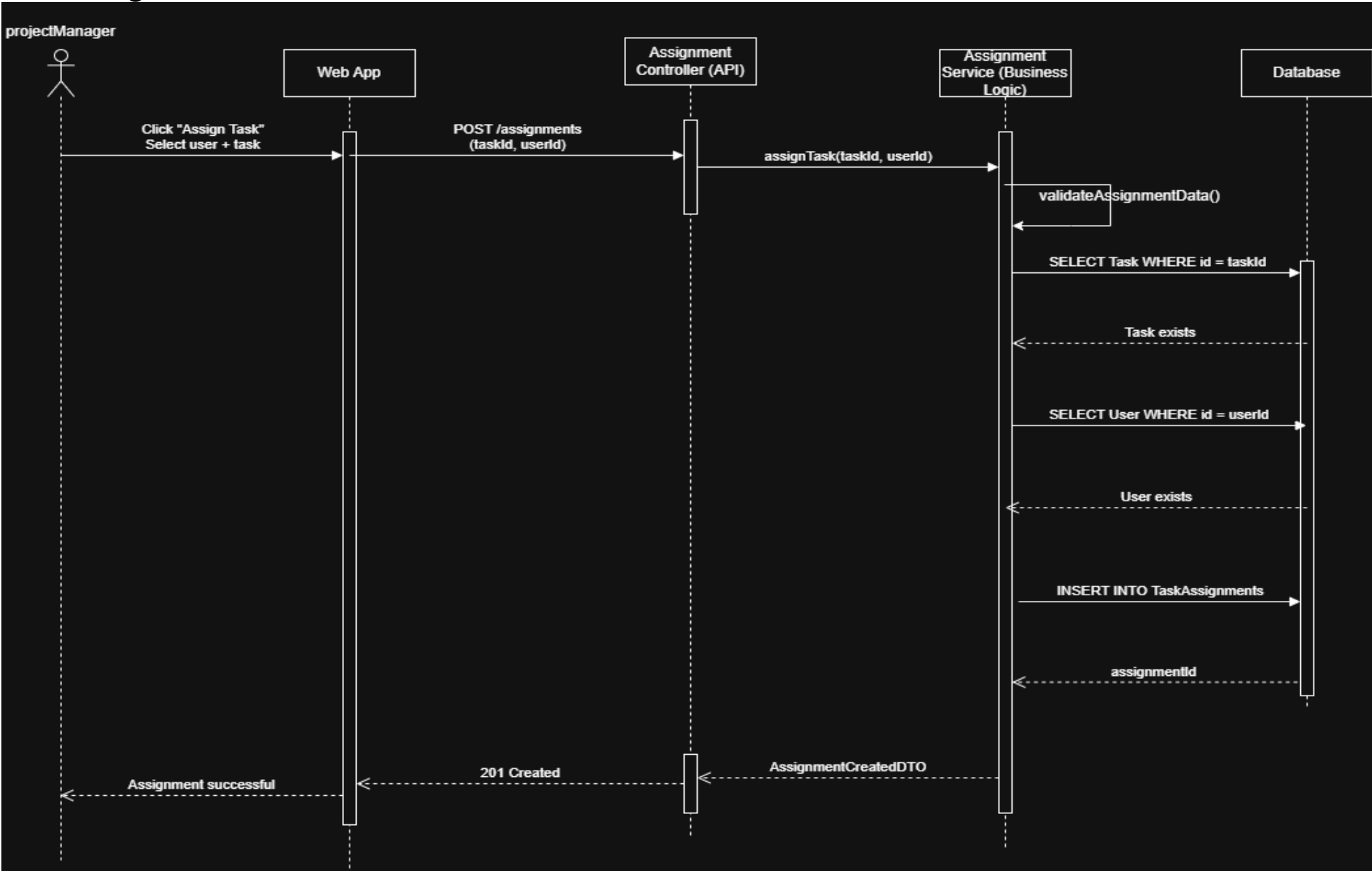
create project



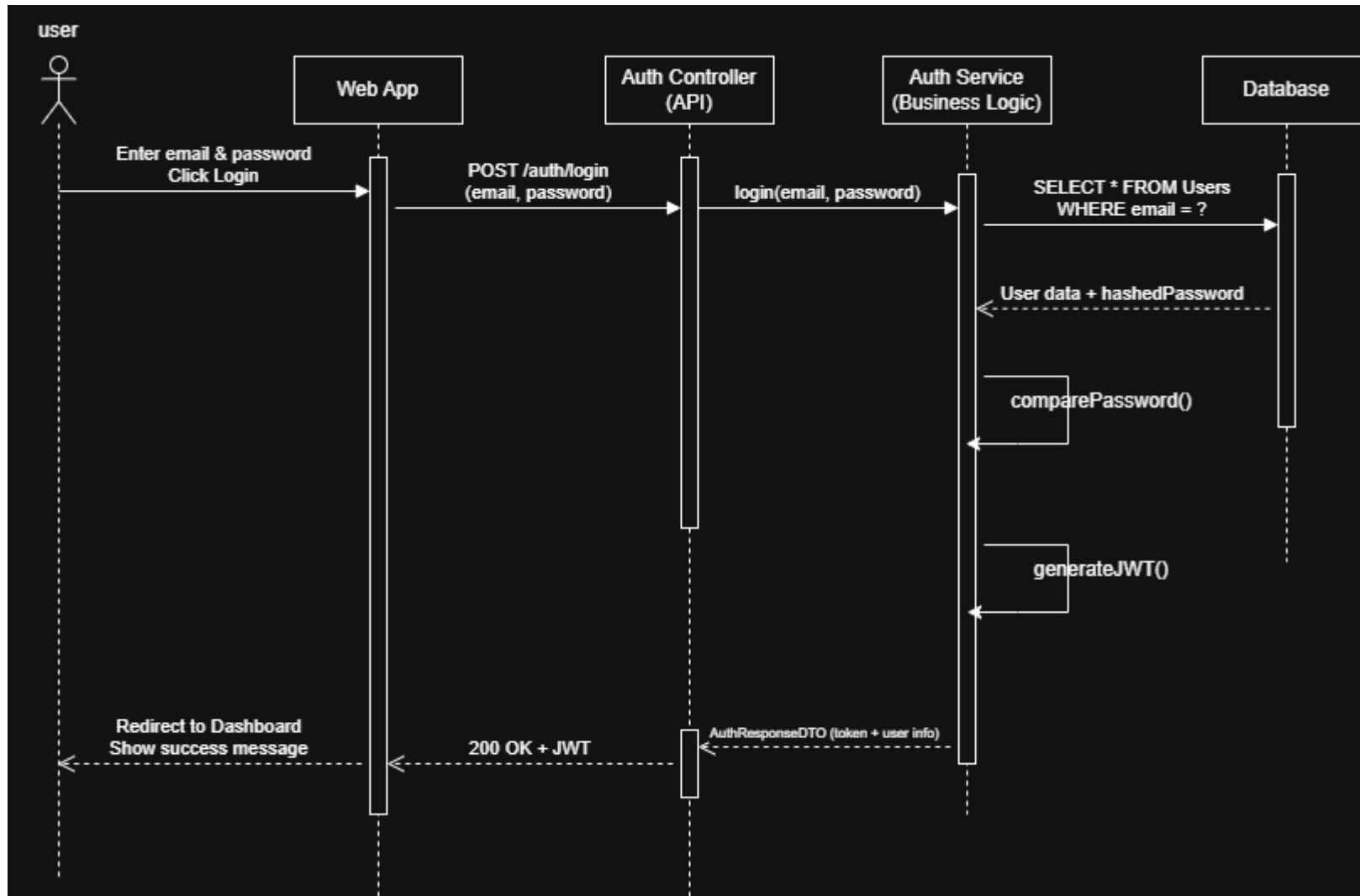
create task



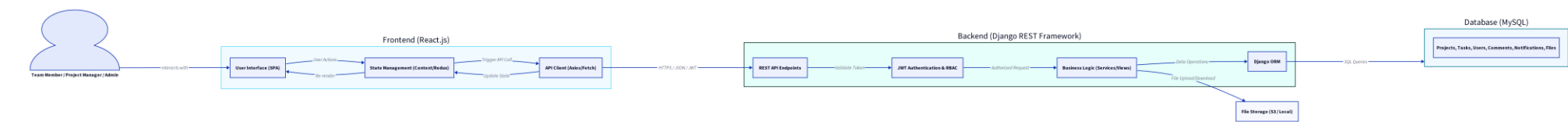
task assign



login sequence



agileflow architecture



UI Draft

Figma UI